Python - Day - 11

Creating a Tuple

x = (10, 20, 30, 40, 50) → paranthesis

python.exe → machine code

<class Tuple>

Built in datastructure

→

a = ( ) → empty Tuple

a = (10) → <class int>

a = (10) → int

↳ expression → ↳ expression

↳ def (parameter)

a (10,)

a (10,) → <class tuple>

↳

→ ① using paranthesis → () → sa

→ Using Tuple Constructor   list() → Const

Sy

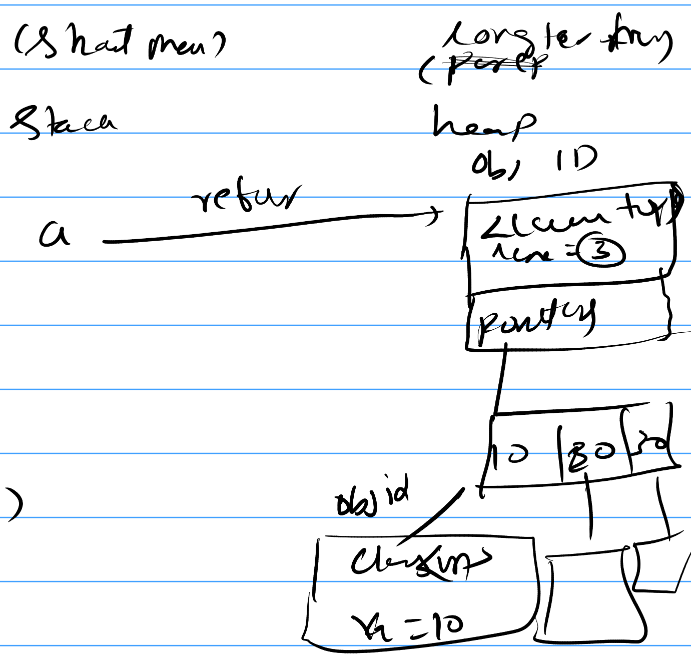
variable = tuple(iterable)

↳

→

a = tuple([10,20,30])
print(a)
o # (10,20,30)
↳ for) iterable
L

o a = (10,20,30)



-> Tuple immutable

-> update
~~a~~ = (10, 20,30)

a[0] -> print(10)

a[0] = 20 ✗

print(a) ↳ Tuple obj not change
(error)

del : python delete -> . Syntax
del variable

a = (10,20,30)

del a

print(a) a not define error

Built in functions :-

① find the length of tuple

① find the len of Tuple:-

→ To find the how many elements present in tuple

→ by using the len()

```
a = (10,20,30)
n = len(a)
print(n)
print(len(a))
```

→ max()

```
a = ((10,20,30))
b = max(a)
print(b)    # 30//

~~print~~a=
```

→ min()

```
→ a= ((10,20,30))
b = ~~max~~(a)
print(b)

#10
```

→ sum →

→ a

```
a = (10,20,30)
b = sum(a) →
print(b)  # 60
```

→ sorted

```
                        → obj, 1D
a = (30,10,20,5)                      # 5,10,20,30
b = sorted(a) → obj, 1D
print(a)       [illegible]  Numeric Item   30,20,10,5
```

Tuple methods:

-> count: Count is a tuple method which use no. of time a element appears Tuple

tuple.count(element)

a = (10, 20, 30, 30, 20, 10)

~~b =~~ a.count(10)

print(b) # ②

-> index: This tuple method which is used find the index of an element that present Tuple

tuple.index(element)

    0 1 2
a = (10, 20, 30)
   -3 -2 -1

b = a.index(20)

print(b) # 1 //

Joining of Tuple:

(T)

-> a = (1, 2) -> obj ID }
b = (3, 4) -> obj ID } +
c = a + b <- obj ID
print(c) # (1, 2, 3, 4)

Imp: Immutable

- Tuple ↑
-> faster than list
-> check for resizing ↑ ✓
-> growth strategy

Hashability -> Immutable

unHashable -> mutable.

a = (10, 20, 10)

b = index(0)

print(b) #